2.8 Actors

Actor	Role in the System
Student / Operator	Runs the application, monitors aircraft and exports data.
PingStation receiver	Provides the structured live receiver stream used by the decoder.
Browser client	Requests Flask APIs and renders the dashboard.
Storage module	Stores current records, trajectory samples and export files.
Future analysis module	Uses saved trajectories for comparative prediction research.

2.9 Use-Case Descriptions

Use Case	Primary Actor	Brief Description
Start tracker	Student / Operator	Launch the server and initialize the decoder, storage and data source.
Replay captured data	System	Read saved PingStation frames and feed them to the decoder in batches.
Decode aircraft record	Decoder	Extract available identity, position and motion fields.
View live aircraft	Browser client	Fetch JSON records and render aircraft markers.
Select aircraft	Student / Operator	Open details, route trace and altitude history.
Save trajectory sample	Storage module	Write meaningful movement samples to SQLite and CSV.
Export trajectory	Student / Operator	Create a trajectory CSV for a chosen time range.
Review logging status	Student / Operator	Check database, matrix CSV and workbook logging status.